

On Curriculum Learning for Commonsense Reasoning

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Abstract

Commonsense reasoning tasks follow a standard paradigm of finetuning pretrained language models on the target task data, where samples are introduced to the model in a random order during training. However, recent research suggests that data order can have a significant impact on the performance of finetuned models for natural language understanding. Hence, we examine the effect of a human-like easy-to-difficult curriculum during finetuning of language models for commonsense reasoning tasks. We use paced curriculum learning to rank data and sample training mini-batches with increasing levels of difficulty from the ranked dataset during finetuning. Further, we investigate the effect of an adaptive curriculum, i.e., the data ranking is dynamically updated during training based on the current state of the learner model. We use a teacher model to measure difficulty of each sample and experiment with three measures based on question answering probability, variability and out-of-distribution. To understand the effectiveness of curriculum learning in various scenarios, we apply it on full model fine-tuning as well as parameter-efficient prompt-tuning settings. Our results show that fixed as well as adaptive curriculum learning significantly improve performance for five commonsense reasoning tasks, i.e., SocialIQA, CosmosQA, CODAH, HellaSwag, WinoGrande in both tuning settings. Further, we find that prioritizing the difficult samples in the tail end of training improves generalization to unseen in-domain data as well as out-of-domain data. Our work provides evidence and encourages research into curriculum learning for commonsense reasoning.¹

1 Introduction

Curriculum learning (Elman, 1993; Bengio et al., 2009) is an alternative to the typical uniform random sampling of training data and is motivated by

¹Code: https://github.com/adymaharana/curriculum_learning

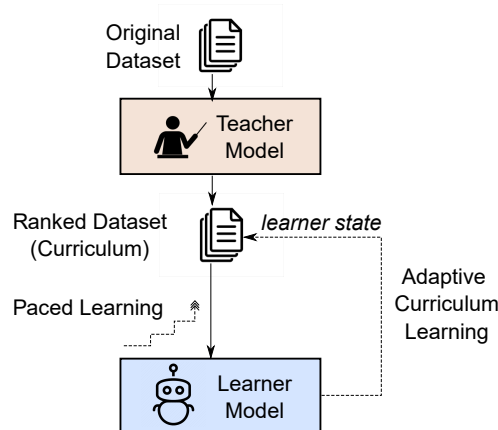


Figure 1: Curriculum Learning. The curriculum of the learner model i.e., the ranked dataset for finetuning is prepared using scores from the teacher model. The ranking remains unchanged in *fixed curriculum learning* whereas the learner state is used as feedback to update the ranking in *adaptive curriculum learning*.

the gradual progression of human learning from easier to difficult concepts (see Figure 1). In the machine learning paradigm, a ‘teacher’ ranks the training samples from easy to difficult and introduces them to the ‘learner’ in that order. Dodge et al. (2020) show that randomly initialized training data orders can lead to large variance in model performance on the GLUE benchmark (Wang et al., 2018). In light of such evidence, we seek to answer: Does a meaningful data order such as a curriculum based on model confidence or dataset distribution outperform a random data order? Such experiments have been carried out for some NLP tasks like machine translation (Platanios et al., 2019) and natural language understanding (Xu et al., 2020) with positive outcomes. While large pretrained models (PTLMs) have been achieving high performance on such tasks, their commonsense reasoning abilities have been limited. Moreover, the process of commonsense acquisition in humans has been shown to be informative for developing algorithms to accomplish the same in machines (Zhu et al., 2020). Hence, we study the effect of a human-like curricu-

lum learning to improve the finetuning of PTLMs for commonsense reasoning tasks.

To impose structure on the data order for sampling training mini-batches, we adopt paced curriculum learning by transfer as proposed in [Hacohen and Weinshall \(2019\)](#). In this method, a pacing function determines the speed at which the ranked data is introduced to the model during training. Ranking of the training dataset is performed using outputs from a pretrained network which has been finetuned on the target dataset using a random training order. We refer to this approach as *fixed curriculum learning*. During human acquisition of skill sets, a student can benefit from a curriculum that is continuously adjusted by the teacher according to the learning progress of the student. Hence, we also investigate *adaptive curriculum learning* for commonsense reasoning tasks. The initial data order imposed by the teacher model is updated at regular intervals during training by taking the learner model’s current state into account ([Kong et al., 2021](#)). Importantly, we propose to reverse the ranking to a difficulty-to-easy curriculum in ACL, in order to reinforce feedback from the hard-to-learn data points, which has been shown to be beneficial for generalization ([Swayamdipta et al., 2020](#)). In order to measure difficulty, we explore three different data-sample informativeness scoring methods i.e. Question Answering Probability (QAP) ([Zhang and Bansal, 2019](#)), Energy-based Out-of-Distribution Score ([Liu et al., 2020](#)) and Cartography-based Variability ([Swayamdipta et al., 2020](#)). Our work is most related to [Xu et al. \(2020\)](#) which splits the training data into N meta-datasets, trains N models for computing the curriculum and follows a heuristically designed training regimen. In contrast, we train a single model for computing the curriculum and use Bayesian optimization ([Snoek et al., 2012](#)) to find the best pacing of the curriculum for the target dataset, which is more effective than [Xu et al. \(2020\)](#) as we show in Sec. 5.4, besides being computationally efficient.

We analyze these methods on five commonsense reasoning datasets dealing with various tasks such as reasoning about social interactions (SocialIQA; [Sap et al. \(2019\)](#)), reading comprehension (CosmosQA; [Huang et al. \(2019\)](#)), natural language inference (HellaSwag; [Zellers et al. \(2019\)](#)), pronoun resolution (WinoGrande; [Sakaguchi et al. \(2020\)](#)) and adversarial commonsense (CODAH; [Chen et al. \(2019\)](#)). We explore curriculum learning in full-

model finetuning as parameter-efficient tuning and show significant improvements using curriculum learning on each of these datasets. We also demonstrate that curriculum learning prevents the learner model from over-fitting on the training set, which leads to improved generalization to in-domain and out-of-domain data.

2 Related Work

Curriculum learning (CL) is widely used in reinforcement learning ([Zaremba and Sutskever, 2014](#); [Matiisen et al., 2019](#); [Graves et al., 2017](#)) and neural machine translation ([Platanios et al., 2019](#); [Kocmi and Bojar, 2017](#); [Guo et al., 2020](#)). [Sachan and Xing \(2018\)](#), [Penha and Hauff \(2020\)](#), [Xu et al. \(2020\)](#) and [Jafarpour et al. \(2021\)](#) demonstrate the effectiveness of CL for question generation, information retrieval, natural language understanding and named entity recognition respectively. To the best of our knowledge, we are the first to examine the efficacy of CL for commonsense reasoning.

Various task-specific measures for sample complexity have been proposed in previous works, such as inter-annotator agreement for natural language inference ([Laverghetta Jr et al., 2020](#)), sub-graph depth for Abstract Meaning Representation (AMR) structures ([Wang et al., 2021a](#)), noise rate in PCA jittering-based data augmentation ([Ye et al., 2021](#)), sentence length for sequence modelling ([Cirik et al., 2016](#)), and semantic similarity for sentiment analysis ([Han and Myaeng, 2017](#)) etc. [Wang et al. \(2021b\)](#) use a neural density estimator to model graph embeddings distribution for the task of graph classification. We use scores based on pretrained models ([Zhang and Bansal, 2019](#); [Swayamdipta et al., 2020](#); [Liu et al., 2020](#)) to compute difficulty.

[Kong et al. \(2021\)](#) and [Cai et al. \(2020\)](#) demonstrate that an adaptive curriculum can improve convergence for image classification and neural response generation respectively. We examine adaptive CL for commonsense reasoning.

3 Methods

3.1 Curriculum Learning

Curriculum learning has two axes of variations, i.e., ranking of samples in terms of difficulty, and transitioning of easy to difficult samples during training. Following the transfer method proposed by [Weinshall et al. \(2018\)](#), we use the predictions of a model that has been trained on target dataset (without curriculum learning) in order to rank the synthetic as

well as original samples by difficulty (see *Teacher Model* in Fig. 1). We adapt the fixed pacing function (Hacohen and Weinshall, 2019) to implement transition of easy to difficult examples and optimize for hyperparameters of the pacing functions using Bayesian optimization. The pacing function $p_f(i)$ is used to determine a sequence of subsets $X_1, \dots, X_m \subseteq X$ of size $|X_i| = p_f(i) * |X|$ from which mini-batches $\{B_i\}$ are sampled uniformly during training. Here, X is the ranked training dataset. The fixed pacing function is comprised of three parameters: (1) starting percentage, (2) increase factor, and (3) step length. The number of training iterations in each step of curriculum learning is defined as step length. The starting percentage decides maximum difficulty of the training samples introduced to the model in first step of curriculum learning. The increase factor is used to exponentially scale up the maximum difficulty at the end of each step. The usage percentage is calculated as $p_f(i) = t * \lambda^{\lfloor i/S \rfloor}$ where t, λ, S and i are the tunable parameters i.e. starting percent, increase factor, step length and current training iteration, resp. Training is initialized by sampling from $t\%$ of the ranked dataset and the usage percentage is re-computed at the end of every S iterations.

3.2 Adaptive Curriculum Learning

When the optimal curriculum for a student is not known in advance, a teacher usually draws up a curriculum based on past teaching experience and then adjusts with the learning progress of the student. Accordingly, Kong et al. (2021) propose initializing the curriculum using the difficulty score obtained from the teacher model and then adapt the score to the current state of the learner model (see Fig. 1). During training, the scores are updated after every L optimization steps. At the $(k+1)$ th update, the difficulty score is computed as $\mu_{k+1} = (1 - \alpha)\mu_k + \alpha\mu_{cur}$ where μ_{cur}, μ_k and μ_{k+1} are difficulty scores after the current step, the k th and the $(k+1)$ th updates respectively. α and L are tunable hyper-parameters. The ranking is *flipped to a difficult-to-easy curriculum* using the updated scores for subsequent training, in order to maximize exposure to difficult samples.

3.3 Difficulty Scoring Functions

Question Answering Probability (QAP). The probability that the teacher model can correctly predict the answer to a question is a measure of model confidence for that particular data sample (Zhang

and Bansal, 2019). We propose to use this metric to rank datasets i.e. data samples with high QAP are considered as easy and those with low QAP are treated as difficult examples. Given a model with parameters θ , the QAP μ_i for question-answer pair (x_i, y_i^*) is measured as $\mu_i = p_\theta(y_i^*|x_i)$.

Model Variability. Swayamdipta et al. (2020) propose the model confidence ($\hat{\mu}_i$) and variability ($\hat{\sigma}_i$) measures to identify the effect of data samples on the model’s generalization error. Specifically, $\hat{\mu}_i = \frac{1}{E} \sum_{e=1}^E p_\theta(y_i^*|x_i)$ and $\hat{\sigma}_i = \sqrt{\frac{\sum_{e=1}^E (p_\theta(y_i^*|x_i) - \hat{\mu}_i)^2}{E}}$, where E is the number of training epochs. We rank samples in the ascending order of variability i.e. samples with low-variability are ranked as easy.

Energy. Liu et al. (2020) show that the energy score can be reliably used for distinguishing between in- and out-of-distribution (OOD) samples. We use this metric to rank OOD samples as "difficult" and in-distribution samples as "easy" in our curriculum. Energy of a given sample is computed as $E = -T * \log \sum_i^K \exp^{f_i(x)/T}$ where $f_i(x)$ are the logits for a given sample, taken from the teacher model, and T is the temperature.

4 Experimental Setup

We use a suite of large and small datasets formatted as multiple-choice question answering tasks for our experiments. SocialQA (Sap et al., 2019), CosmosQA (Huang et al., 2019) and WinoGrande-XL (Sakaguchi et al., 2020) contain upto 60K training samples. Additionally, we use the CODAH dataset (Chen et al., 2019) and also follow the method in Yang et al. (2020) to create HellaSwag-2K (Zellers et al., 2019) for testing our methods in low-resource scenarios. Models are evaluated using the respective task-specific accuracies (see Appendix for dataset statistics). We use pretrained RoBERTa_{LARGE} (Liu et al., 2019) as the teacher as well as learner model in our full-model finetuning experiments. In the first stage, the teacher model is finetuned on randomly sampled training data and used to compute the difficulty scores. Models are also subjected to grid-search based tuning of training hyperparameters in the first stage whenever necessary. In the second stage, we use Bayesian optimization for finding the optimum pacing function parameters for fixed as well as adaptive CL (and use the same training hyperparameters as first stage). For adaptive learning, we set $L = S$ and

Method	SIQA	CosQA	CDH	H2K	WG
<i>Results on test set</i>					
RoBERTa	76.74*	79.23*	82.32	73.40	79.12
+ fixed CL	78.14	80.04	83.91	75.42	79.51
+ adaptive CL	78.53	80.43	84.75	76.10	79.97
<i>Results on validation set</i>					
RoBERTa	77.78	80.45	84.28	74.72	79.63
+ fixed CL	78.86	81.0	85.92	77.23	80.11
+ adaptive CL	79.22	81.57	86.03	77.89	81.05

Table 1: Results on commonsense datasets using RoBERTa and various CL methods. *values are taken from leaderboards. (SIQA = SocialIQA, CosQA = CosmosQA, CDH = Codah, H2K = HellaSwag2K, WG=WinoGrande-XL)

Method	SIQA	CosQA	CDH	H2K
RoBERTa	77.78	80.45	84.24	76.80
+ QAP	78.86	81.69	85.92	77.98
+ Energy	77.93	80.67	85.09	75.94
+ Variability	78.41	81.58	84.95	77.81

Table 2: Ablation results on validation set of commonsense reasoning datasets using fixed CL with RoBERTa and various difficulty measures.

optimize for α , t , λ and S parameters. For prefix-tuning experiments, we use GPT2_{LARGE} (Radford et al., 2019) as the PTLM for tuning. See Appendix for dataset statistics and hyperparameter bounds.

5 Results & Analysis

5.1 Main Results

Our experiments with curriculum learning yield upto 2% improvements across five commonsense reasoning tasks using RoBERTa model (see Table 1). Fixed CL results in 1.4%, 0.81% and 0.4% improvement over baseline i.e. no CL (see row 1 in Table 1) for the larger datasets SocialIQA, CosmosQA and WinoGrande respectively. With adaptive CL, the improvements increase to 1.79%, 1.20% and 0.85% respectively. For the smaller datasets, we see similar benefits i.e. 1.7% and 2.5% with fixed CL, and 2.4% and 3.17% with adaptive CL over the baselines of Codah and HellaSwag-2K respectively. These results are obtained using QAP as difficulty score, which is the superior metric for measuring sample difficulty in multiple-choice datasets (see Sec. 5.2). The optimum α values in our experiments are closer to 1.0, which suggests that the initial ranking using teacher model is not quite useful after the first S training steps.

To investigate the effectiveness of curriculum learning, we compare the difficulty scores on the training and validation sets for the teacher vs. learner models. See Figure 4 for a demonstration

Method	WG (ID)	WSC (OOD)
RoBERTa	79.63	88.07
+ fixed CL	80.11	88.77
+ adaptive CL	81.05	90.17

Table 3: In-domain (ID) and out-of-domain (OOD) accuracies for RoBERTa_{LARGE} models trained with and without CL on the WinoGrande-XL (WG) dataset.

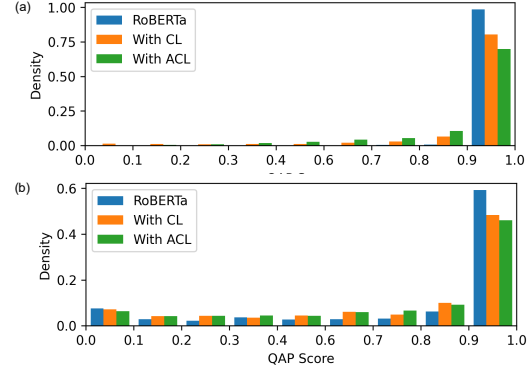


Figure 2: Comparison of QAP scores on the (a) training and (b) validation sets of HellaSwag-2K, for the RoBERTa models trained with and without CL.

of the same for HellaSWAG-2K. We observe that the learner model is less confident about the training data implying that there is less overfitting. This results in improved generalization to unseen data in the validation set, and more uniform distribution of QAP scores over the samples. We observe similar trends in other datasets as well (see Appendix for more figures), which suggests that curriculum learning acts as a regularizer during training. In order to further test this hypothesis, we evaluate the out-of-distribution generalization of WinoGrande models by evaluating on the Winograd Schema Challenge (WSC) dataset (Levesque et al., 2012). We observe 0.7% and 2.1% improvement in the performance on WSC using fixed and adaptive CL respectively (see Table 3), indicating that adaptive CL is especially effective at promoting generalization of the trained model. This result also aligns with the finding in Swayamdipta et al. (2020) that hard-to-learn examples play a significant role in learning and generalization.

5.2 Difficulty Metrics

We use three measures of difficulty for ranking the samples in the training data. Results on fixed CL with RoBERTa are shown in Table 2. We see the largest improvements with QAP, similar behavior with variability, and lesser or no improvements with energy. Scatter plot between QAP and variability for SIQA data samples reveals that samples with higher QAP also tend to have lower variance, hence

Method	MNLI	QNLI	QQP	RTE	SST-2	MRPC	CoLA	STS-B	Avg.
BERT _{LARGE}	86.7	92.5	91.2	76.1	94.0	91.4	66.1	90.2	83.7
+ CL (Xu et al., 2020)	86.6	92.8	91.8	76.2	94.2	91.9	66.8	90.6	84.1
+ fixed CL (ours)	86.8	93.1	91.8	77.1	94.6	92.3	66.8	91.0	85.6
+ adaptive CL (ours)	87.9	93.5	92.7	77.9	94.6	92.4	66.5	91.8	86.1

Table 4: Results on validation sets of the GLUE dataset using fixed and adaptive CL with BERT_{LARGE}.

Method	SIQA	CosQA	CDH	H2K
Prefix-tuning	65.39	68.34	73.16	70.24
+ fixed CL	66.18	69.10	75.28	72.58
+ adaptive CL	66.91	69.42	75.56	72.68

Table 5: Results on validation set of commonsense reasoning datasets using prefix-tuning of GPT2 and curriculum learning with QAP as difficulty measure.

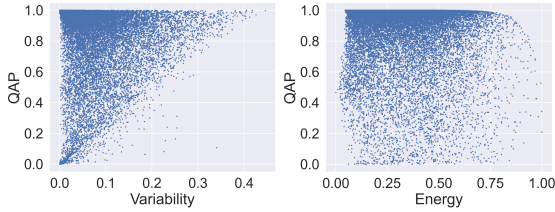


Figure 3: Visualization of scatter plots for QAP vs. Variability (left) and QAP vs. Energy (right) scores from the RoBERTa teacher model for SocialQA.

leading to similar behavior from both metrics in curriculum learning. The energy score fails to yield benefits for CL because the datasets used in our experiments are mostly homogeneous.

5.3 Curriculum Learning for Prefix-Tuning

In the interest of parameter-efficient methods for training PTLMs to perform specialized tasks, we conduct CL experiments for prefix-tuning (Li and Liang, 2021) of GPT2_{LARGE} models on target datasets (see Appendix). We introduce a prefix of length 16 and train a reparameterization network that updates 0.2% of GPT2’s parameters for commonsense-based question answering. Results in Table 5 show that CL yields up to 1.5% improvements for smaller datasets and up to 1% improvement on larger datasets, suggesting that CL could be also effective for prompt-tuning settings.

5.4 Curriculum learning for Natural Language Understanding

We evaluate the performance of our proposed curriculum learning methods on conventional NLU tasks i.e. GLUE. In order to facilitate direct comparison to the meta-dataset approach presented in Xu et al. (2020), we train BERT_{LARGE} models with our fixed as well as adaptive curriculum learning methods on all sub-tasks in GLUE (Wang et al., 2018). Results on the validation sets of GLUE are

SIQA	Cameron gave Casey a drink. He loved helping kids and giving them things. How would Casey feel as a result? [A] very apathetic [B] very grateful [C] somewhat indifferent
Codah	We organized a bonfire party. I [A] brought marshmallows to toast [B] like to play with fire [C] threw a bucket of water at the bonfire [D] howled like a wolf.

Table 6: Most difficult samples of SIQA and CODAH as ranked by QAP scores. Labels are marked in green.

presented in Table 4. Our proposed method outperforms Xu et al. (2020) across all sub-tasks. We see upto 1% improvement on the larger datasets i.e. MNLI, QNLI and QQP. For the smaller GLUE datasets, we observe small improvement margins with curriculum learning approaches, as opposed to the larger improvements for small commonsense reasoning datasets as seen in Table 1.

6 Limitations

Our results are limited to the task of multiple choice question-answering based commonsense reasoning and natural language understanding, but are encouraging and warrant further research into effective adaptive CL methodologies for other NLP tasks. The adaptive curriculum learning method proposed in our paper is expensive for larger datasets since the ranking needs to be recomputed several times during training using the current version of the learner model. Further work is needed to optimize curriculum learning methods for larger datasets like AbductiveNLI (Bhagavatula et al., 2019) and for using larger models like UnifiedQA (Khashabi et al., 2020) for finding the right curriculum.

7 Conclusion

We conduct an empirical analysis of fixed and adaptive curriculum learning (CL) for five commonsense reasoning tasks using pretrained language models (PTLMs). Results show that CL can benefit downstream task performance when introducing a new task to a PTLT, in fully-finetuned as well prompt-tuned settings, and for in-domain as well as out-of-domain data. Our work motivates future research into CL for commonsense reasoning.

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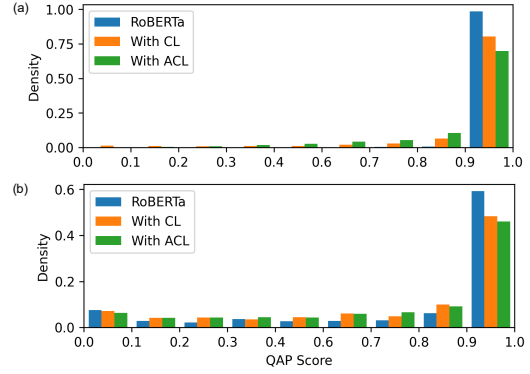


Figure 4: Comparison of QAP scores on the (a) training and (b) validation sets of SocialIQA, for the RoBERTa models trained with and without CL.

A Prefix-Tuning Model

For this scenario, we finetune a $GPT2_{LARGE}$ model for the task of multiple-choice reasoning using prefix-tuning, wherein the weights of pretrained GPT2 are frozen, and only a matrix of prefix embeddings are trained for the target task. Following Li and Liang (2021), we prepend a prefix to the inputs of GPT2 model i.e. $z = [PREFIX, x, y]$ where (x, y) is the question-answer input pair. We also a parameterization network to compute the prefix embeddings i.e., $P_\theta = MLP(P_{\theta'})$ where P_θ is the prefix embedding used during inference, $P_{\theta'}$ is a smaller prefix matrix used for reparameterization during training and MLP is the dense layer used as the reparameterization network. We set the prefix length to 16, and the dimensions of $P_{\theta'}$ to 512. The dimensions of P_θ is 1024, according to the hidden size of $GPT2_{LARGE}$. This leads to a total of 0.2% trainable parameters in our prefix-tuning based GPT2 model.

B Experimental Setup

Datasets. We use two large multiple-choice commonsense-based question answering datasets for our experiments i.e. SocialIQA (Sap et al., 2019) and CosmosQA (Huang et al., 2019). Additionally, we also use the CODAH dataset (Chen et al., 2019) folds released in Yang et al. (2019) and follow their method to create HellaSwag-2K (Zellers et al., 2019) for testing our methods on low-resource scenario. All of the above datasets are being for their intended purposes i.e. research only, in our work. All of these datasets are in the English language. Models are evaluated using the respective task-specific accuracies. See Table 9 for dataset statistics.

Hyperparameter	SocialIQA	CosmosQA	CODAH	HellaSwag2K	WinoGrande (XL)
Starting Percent (sp)	0.26	0.37	0.5	0.26	0.39
Increase Factor (inc)	1.89	1.74	1.32	1.05	1.8
Update Steps (S)	1789	1745	332	199	405
Adaptive Factor (α)	0.93	0.99	0.55	0.09	0.19

Table 7: Best adaptive curriculum learning pacing function parameters for finetuning RoBERTa on various datasets

Parameter	Bounds
Grid-search Based Optimization	
Learning Rate	[1e-5, 5e-5, 1e-6, 5e-6]
Batch Size (inc)	[8, 16]
Total Train Epochs	[3, 4, 5]
Bayesian Optimization	
Starting Percent (t)	[0.01, 0.5]
Increase Factor (λ)	[1.05, 2.0]
*Update Steps (S)	[0.01, 3]*250
Adaptive Factor (α)	[0, 1.0]

Table 8: Optimization bounds for grid-search based tuning of training hyperparameters and bayesian optimization of curriculum learning parameters. *We expand the search window for update steps to 2500 for larger datasets in adaptive CL

Split	SIQA	CosQA	CDH	H2K	WG
Train	33,410	25,778	1666	2000	40,498
Dev	1954	3000	566	1000	1266
Test	2224	7000	566	5000	1767

Table 9: Number of samples in each split of the datasets used in our experiments. (SIQA = SocialIQA, CosQA = CosmosQA, CDH = Codah, H2K = HellaSwag2K, WG=WinoGrande-XL)

Models & Hyper-parameters. We use RoBERTa_{LARGE} Liu et al. (2019) as the main PTLM in our experiments. We perform Bayesian optimization for finding the optimum parameters for fixed as well as adaptive CL. For adaptive learning, we set $L = S$ and optimize for α , t , λ and S parameters. All models are also subjected to grid-search based tuning of training hyperparameters.

B.1 Hyperparameter Tuning Bounds

We perform grid-search based optimization for finding the best training hyperparameters for each training scenario and dataset. For curriculum learning based parameters, we first perform bayesian optimization (Snoek et al., 2012) of curriculum learning parameters using the best training hyperparameters for the baseline (non-CL) model. For bayesian optimization, each model is subjected to 15 iterations of optimization with 3 restarts and the tuning



Figure 5: Demonstration of the optimized learning pace of RoBERTa for various datasets. Usage percentage k (y-axis) at training step n (x-axis) refers to the top $k\%$ easiest examples of the ranked dataset.

is based on accuracy on validation set.

For the prefix-tuning models, we perform grid-search based tuning of training hyperparameters according to the bounds reported in Table 8 and in addition, up to 10 training epochs.

Hyperparameter	SocialQA	CosmosQA	CODAH	HellaSwag2K	WinoGrande (XL)
Learning Rate	5e-6	5e-6	1e-5	1e-5	1e-5
Epochs	3	4	5	5	5
Max Gradient Norm	1.0	1.0	1.0	1.0	1.0
Weight Decay	0.01	0.01	0.01	0.01	0.01
Batch Size	8	8	16	16	16
Max Length	128	128	90	128	70
Warmup Ratio	0.06	0.06	0.06	0.06	0.06
LR Decay	Linear	Linear	Linear	Linear	Linear
Optimizer	AdamW	AdamW	AdamW	AdamW	AdamW
Hardware	RTX 2080 Ti	RTX 2080 Ti	RTX 2080Ti	RTX 2080Ti	RTX 2080Ti
Single GPU training time	5 hours	5 hours	2 hours*	2 hours	5 hours

Table 10: Training hyperparameters for finetuning RoBERTa on various datasets.